

A Comparative NLP Framework for Mental Health Text Classification

Goal: This project aims to develop an NLP model that can identify and classify mental health conditions based on text. Using the **Sentiment Analysis for Mental Health** dataset from Hugging Face, which contains user-generated statements labeled into seven categories—Depression, Anxiety, Stress, Suicidal, Bipolar, Personality Disorder, and Normal—the model will analyze linguistic patterns to detect emotional and psychological cues. The motivation stems from the need for early detection of mental health issues expressed through online platforms. By comparing traditional machine learning, Word2Vec-based neural networks, and transformer models like BERT, the study seeks to determine which approach most effectively captures the nuanced language associated with mental health expressions.

Proposed Methodology

The proposed methodology follows a step-by-step experimental workflow combining text preprocessing, model development, and comparative evaluation.

1. Data Collection and Understanding

The dataset `btwitssayan/sentiment-analysis-for-mental-health` from Hugging Face will be used. Each entry contains a short text (self-expression) and a corresponding mental health label.

2. Data Preprocessing

Convert all text to lowercase, remove URLs, punctuation, emojis, and stopwords. Tokenize sentences and perform lemmatization. Split the dataset into 80% training and 20% testing. Encode categorical labels numerically for model compatibility.

3. Modeling Approaches

The project will implement and compare three distinct approaches:

(a) Traditional Word-Based Model (TF-IDF + Multinomial Naive Bayes):

A probabilistic baseline model leveraging word frequency and term importance. It provides a quick, interpretable foundation for text classification.

(b) Word-Embedding-Based Model (Pre-trained Word2Vec Embeddings + Feed-Forward Neural Network):

Each word will be represented as a dense vector using semantic embeddings. Sentence vectors will be computed by averaging token embeddings and fed into a small neural network for classification.

(c) Transformer-Based Model (Fine-Tuned DistilBERT (or MentalBERT)):

A pre-trained transformer model will be fine-tuned on the dataset to capture deep contextual and emotional semantics. This model is expected to perform best in detecting nuanced psychological cues.

Evaluation Methodology

The models will be compared across quantitative and qualitative metrics to assess their predictive capability and interpretability.

(a) Performance Metrics

- Accuracy: Overall proportion of correctly classified samples.
- Precision, Recall, F1-Score: To account for class imbalance among different disorders.
- Macro-Averaged F1-Score: Ensures equal weight for all classes.
- Confusion Matrix: To visualize model strengths and weaknesses across categories.

(b) Cross-Validation

- Employ 5-fold cross-validation to reduce bias and estimate model generalizability.

(c) Cross-Lingual Evaluation

- Translate a subset of the dataset (e.g., English → French) using an open-source translation model to test multilingual generalization of the fine-tuned transformer.
- Evaluate if contextual meaning and emotion mapping remain consistent across languages.

Timeline

Week	Task
Week 1	Dataset exploration, cleaning, and preprocessing
Week 2	Implementation of TF-IDF + Multinomial Naive Bayes baseline
Week 3	Word2Vec embedding generation and neural network training
Week 4	Fine-tuning DistilBERT (or MentalBERT) model
Week 5	Model evaluation, cross-lingual testing, and visualization

References:

1. Sayan, B. T. W. I. T. (2024). Sentiment Analysis for Mental Health [Dataset]. Hugging Face (<https://huggingface.co/datasets/btwitssayan/sentiment-analysis-for-mental-health>)
2. Tadesse, M. M., Lin, H., Xu, B., & Yang, L. (2019). Detection of Depression-Related Posts in Reddit Social Media Forum. IEEE Transactions on Computational Social Systems, 6(5), 957-968. <https://doi.org/10.1109/TCSS.2019.2918280>
3. Kumar, A., & Garg, A. (2022). Sentiment Analysis for Mental Health Monitoring Using Deep Learning. Journal of Ambient Intelligence and Humanized Computing, 13(4), 1899-1912. <https://doi.org/10.1007/s12652-021-02972-3>